The Order Matters: Path Dependence in Adaptive Rubric-Based LLM Judging

SEBASTIAN TONU[1], IOANA-ALEXANDRA TONU[1],
MARIUS-ALEXANDRU OLARU[2], ȘTEFAN-DANIEL ACHIREI[1]

[1]*Faculty of Automatic Control and Computer Engineering, Gheorghe Asachi Technical University of Iași*
{sebastian.tonu, ioana-alexandra.tonu,
stefan-daniel.achirei}@academic.tuiasi.ro
[2]*Faculty of Computer Science, Alexandru Ioan Cuza University of Iași*
marius-alexandru.olaru@info.uaic.ro

Abstract

Evaluation rubrics can be adapted as new human annotations become available. Whether the resulting criteria depend on the order of these examples, rather than their content alone, remains less clear. We examine this question using HelpSteer2 and Qwen3:14B, comparing a fixed rubric with two approaches to adaptation: updating from a batch of examples or revising the rubric after each example. Sequential updating produces the closest agreement with human ratings (QWK 0.309; Spearman 0.423). This improvement comes with a cost. When the same calibration examples are presented in different orders, sequential rubrics exhibit roughly twice the instability observed with batch updating (0.0637 versus 0.0318). The proportion of scores affected also increases, from 7.0% to 13.6%. Repeated evaluations under identical conditions produce identical scores, making ordinary inference noise an unlikely explanation for these differences. Our results suggest that the history of rubric construction deserves attention alongside agreement with human judgments when assessing the reliability of LLM evaluation methods.

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